JATIN SIKKA

Austin, TX | (919) 345-6263 | jsikka@utexas.edu | jatinsikka.me | linkedin.com/in/jatinsikka | github.com/jatinsikka

EDUCATION

The University of Texas at Austin | M.S. Mechanical Engineering, Robotics **Expected Dec 2027**
Coursework: Human Centered Robotics, Learning-Based Optimal Control, Haptics & Teleoperation, Deep Learning

North Carolina State University | B.S. Mechanical Engineering, Minor in Computer Programming **Dec 2024 | GPA 3.81**

TECHNICAL SKILLS

Software	Python, C++, C, ROS2, MATLAB, Bash; Linux, Git, Docker, AWS, Azure
Robotics & ML	PyTorch, Isaac Lab/Sim, MuJoCo, OpenCV; reinforcement learning (PPO, TRPO), IK, PD control
Mechanical	SolidWorks, CATIA, AutoCAD, FEA, GD&T; 3D printing, laser/waterjet cutting

RESEARCH EXPERIENCE

Human Centered Robotics Lab, UT Austin | PI: Dr. Luis Sentis **Sep 2025 – Present**
Sequent: ticket-to-task humanoid loco-manipulation (Unitree G1, MuJoCo) | github.com/jatinsikka/sequent-g1

- Built a pipeline where a simulated Unitree G1 reads a plant-floor incident ticket, retrieves the matching standard operating procedure (SOP), and executes its steps as walking and manipulation skills in MuJoCo.
- Paired a pretrained whole-body locomotion policy with RL and IK skills to chain walk, pick, carry, place, and press.
- Trained a BERT bi-encoder for SOP retrieval; when it lost to keyword search, switched to a fine-tuned MiniLM that nearly doubled keyword-search accuracy at picking the correct SOP out of 1,012 (58% vs. 30%).

Sim-to-Real and Teleoperation

- Trained PPO reach policies in Isaac Lab and deployed them sim-to-real on a physical SO-101 arm, matching sim joint stiffness and friction to the real servos to remove hardware oscillation.
- Building a Unitree R1 + Revo2-hand Isaac Lab factory scene with Meta Quest teleop to collect manipulation demos.

Intelligent Controls Lab, NC State University | Advisor: Dr. Donggun Lee **Aug 2023 – Oct 2024, Summer 2025**

- Extended TRPO for safe policy updates in autonomous navigation; built a PyBullet + Gym training environment.

PROFESSIONAL EXPERIENCE

Tesla | Software Data Engineering Intern, Body in White (Model Y & Cybertruck) **May 2023 – Aug 2023**

- Engineered a Python + live SQL pipeline automating real-time extraction and transformation of BIW quality metrics.
- Built 10 Tableau dashboards (castings, welds, adhesives, audits) with heatmaps and Pareto charts for defect analysis.
- Cut metrology analysis time 93% by automating MATLAB trend analysis of 74 features (394 raw measurements).

Tesla | Mechanical Engineering Intern, Plastics (Model S & X), Fremont, CA **Jan 2022 – May 2022**

- Designed rocker punch stripper plates in SolidWorks and 3D-printed a prototype, resolving 95% of retraction defects.
- Built a spring-loaded proof-of-concept tool to stop park-assist sensor seating defects on Model S/X.
- Designed and 3D-printed Model 3 frunk clip covers, cutting transport scratching and warping 97%.

Fortive (Hengstler Dynapar) | Project Engineering Intern **Aug 2021 – Dec 2021**

- Led transition of 2 assembly cells (~$500K/month) in a plant acquisition; raised efficiency 33% with custom fixtures.
- Designed encoder and PCB-soldering fixtures (resin-printed, CNC-machined), cutting assembly errors 61% and 31%.

PROJECTS

- **Haptic Teleop (C++ / CHAI3D):** Virtual walls and force fields on a Geomagic Touch; adding force feedback to SO-101.
- **Atlas Whole-Body Control:** QP whole-body controller and DCM walking planner for a simulated humanoid.
- **Vision-Based Sorting Arm (4-DOF):** Developed a camera-guided arm using ML object detection and IK to sort objects.
- **Water-Level Control System (Pentair Senior Design, 1st Place):** Mechatronic design, Raspberry Pi control, ±2 cm.

LEADERSHIP

Senior Design Teaching Assistant, UT Austin | Mentoring 8 capstone teams (robotics, AI, design) **Aug 2025 – Present**